

Introduction & Perception

Introduction to Computer Graphics - Lecture 01

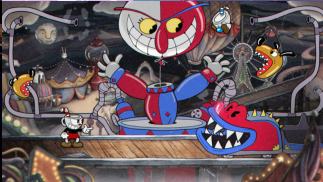
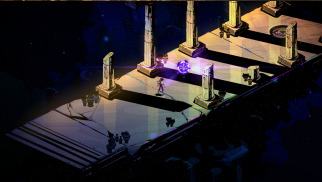
Pirmin Pfeifer

Computer Graphics

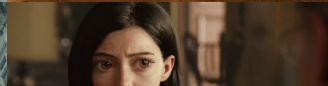
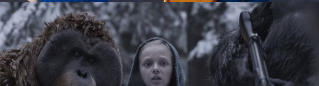
The creation and manipulation of images with a computer



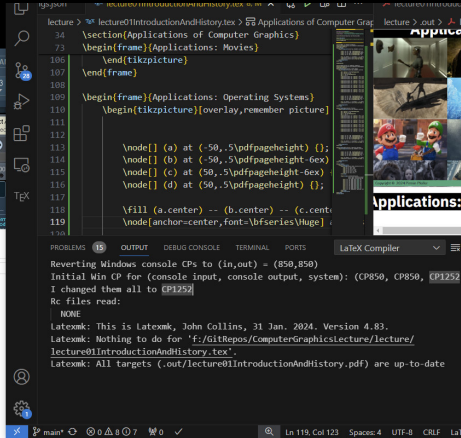
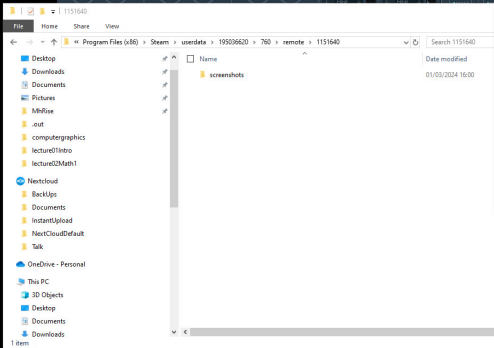
Applications: Video Games



Applications: Movies



Applications: GUI

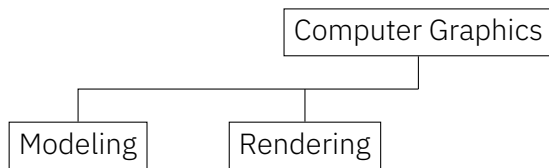


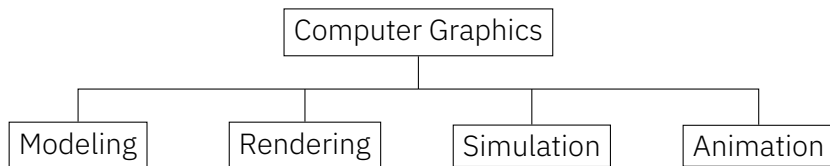
Computer Graphics

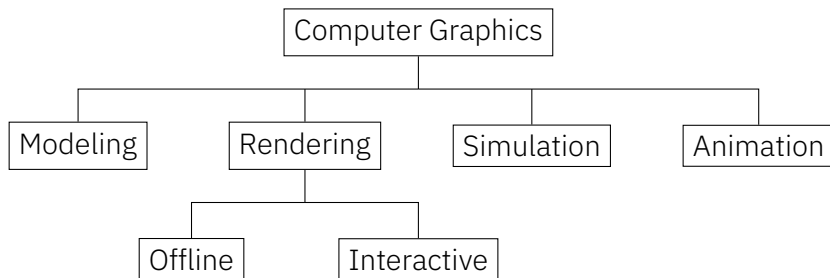
Areas of Computer Graphics

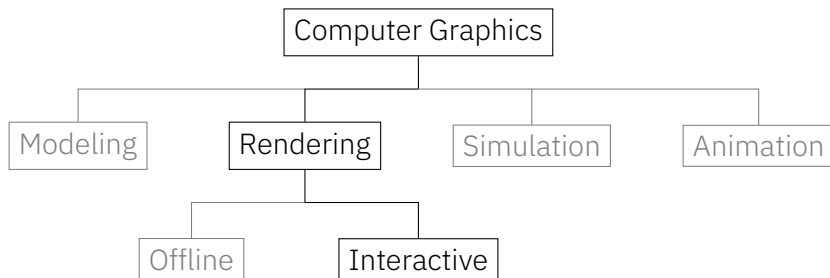


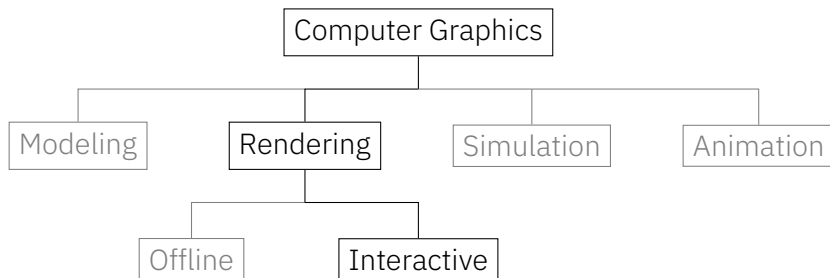
Areas of Computer Graphics









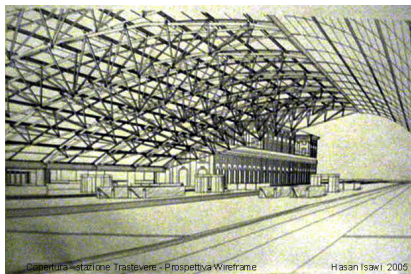


In this lecture we focus on the realtime-rendering aspect of Computer Graphics.

Rendering Question

Given:

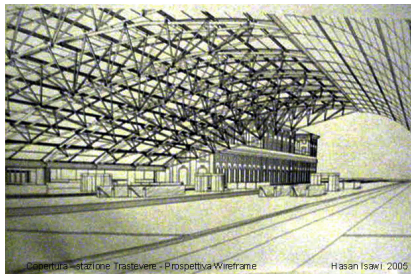
A mathematical representation
of a Scene



Rendering Question

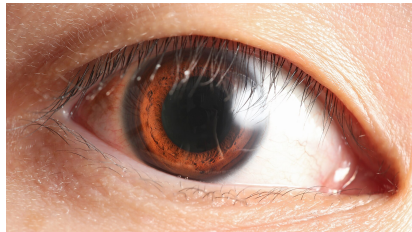
Given:

A mathematical representation
of a Scene

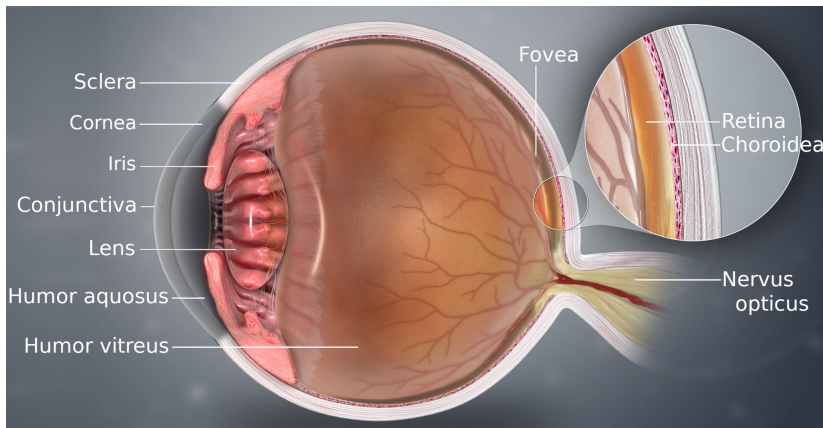


Wanted:

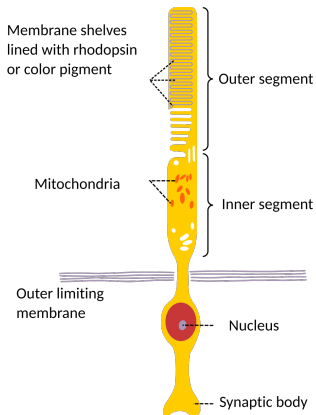
Something to look at (like a
picture)



The Human Eye

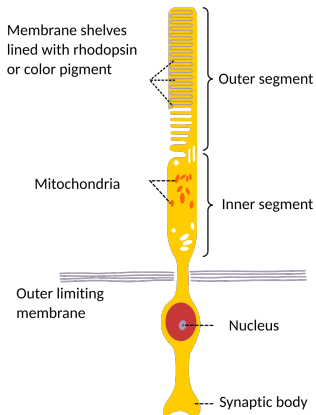


The Human Eye

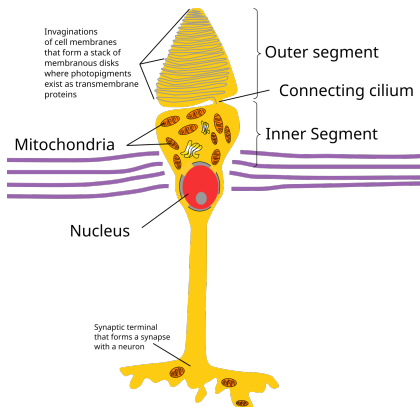


Rod Cell: Primarily perceive
light intensity

The Human Eye

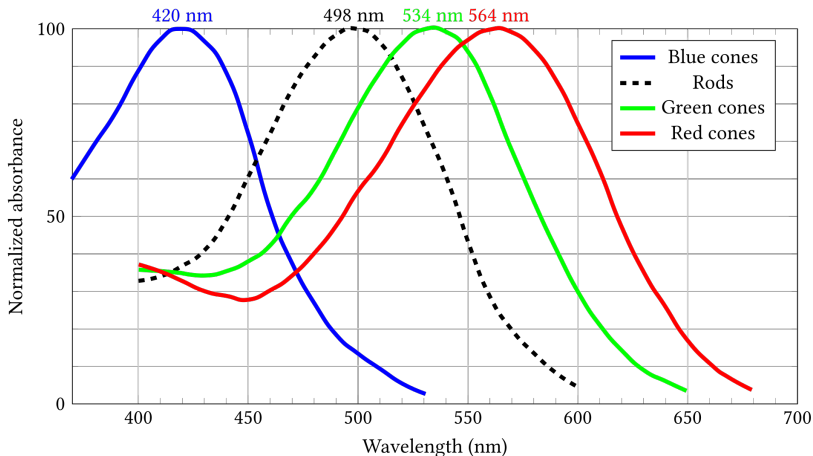


Rod Cell: Primarily perceive light intensity

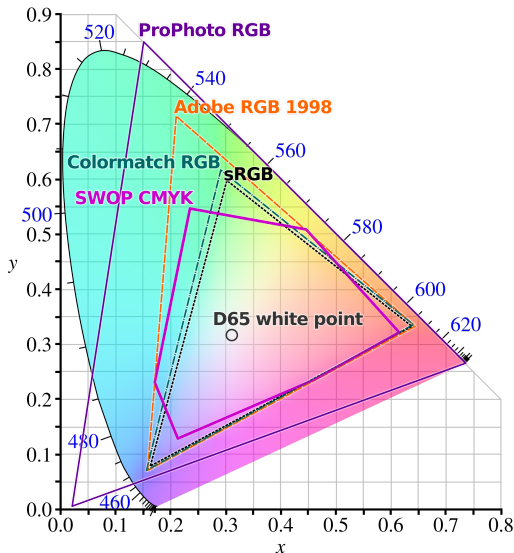


Cone Cell: Primarily perceive colors (blue, green & red)

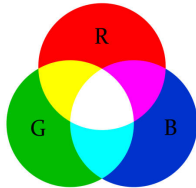
The Human Eye is not equally sensory to all colors:



Our Devices try to replicate the natural light sensations:

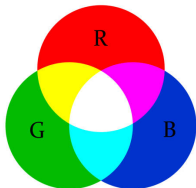


Color Formats: RGB



RGB is an **additive** color format

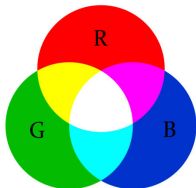
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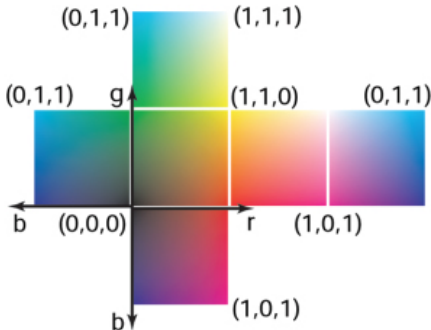
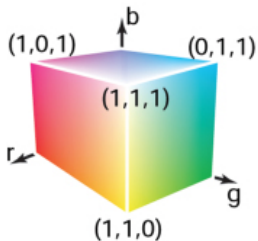
- ▶ Red
- ▶ Green
- ▶ Blue

Color Formats: RGB



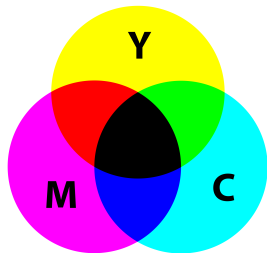
RGB is an **additive** color format with three components:

- ▶ Red
- ▶ Green
- ▶ Blue

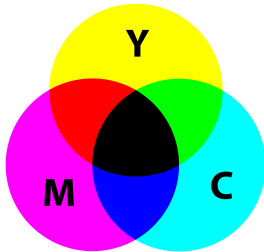


Color Formats: CMY

CMY is an **subtractive** color format



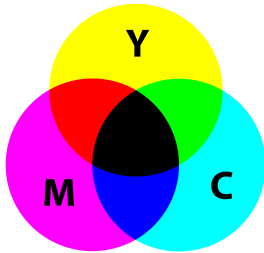
Color Formats: CMY



CMY is an **subtractive** color format

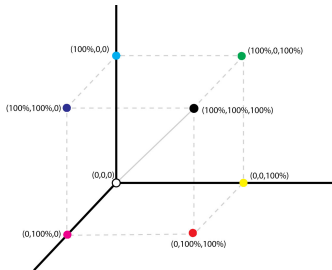
- ▶ Cyan
- ▶ Magenta
- ▶ Yellow

Color Formats: CMY

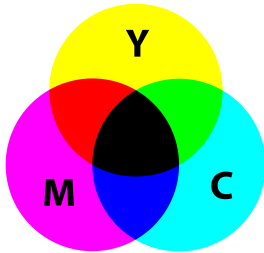


CMY is an **subtractive** color format with three components:

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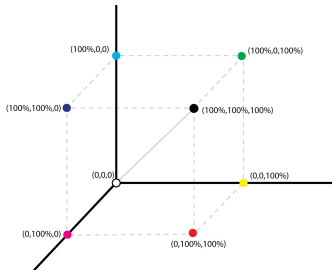
Color Formats: CMY



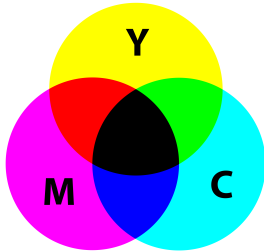
CMY is an **subtractive** color format with three components:

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CMY is mainly used by printers.



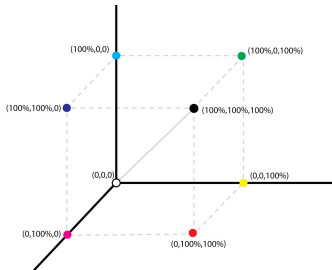
Color Formats: CMY



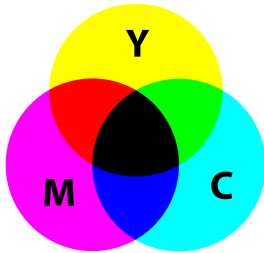
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CMY is mainly used by printers.
CMY and RGB can be converted:



Color Formats: CMY



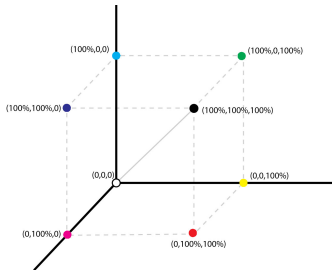
CMY is an **subtractive** color format with three components:

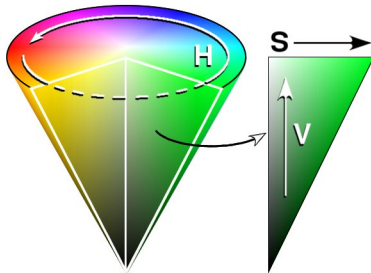
- ▶ Cyan
- ▶ Magenta
- ▶ Yellow

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CMY and RGB can be converted:

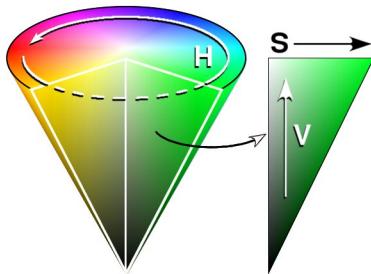
$$RGB = 1 - CMY$$

$$CMY = 1 - RGB$$





Color - HSV



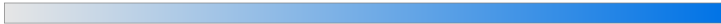
1. Pick a Hue

210



2. Pick a Saturation

100



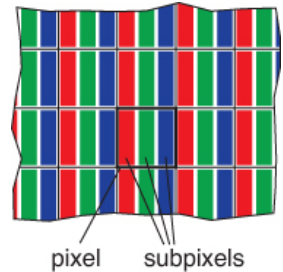
3. Pick a Brightness (37%)

90



Most images are presented on a kind of *raster display*.

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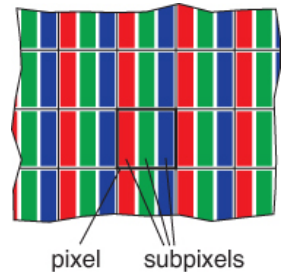


¹Pixel is short for "picture element"

Most images are presented on a kind of *raster display*.

Usually each pixel contains:

- ▶ Red Subpixel
- ▶ Green Subpixel
- ▶ Blue Subpixel

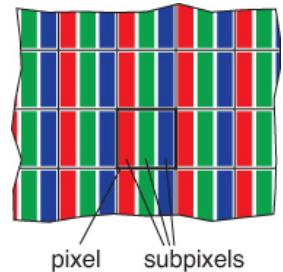


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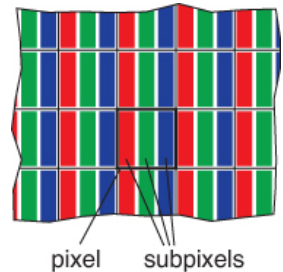
Common Raster Devices:

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Common Raster Devices:

Output:

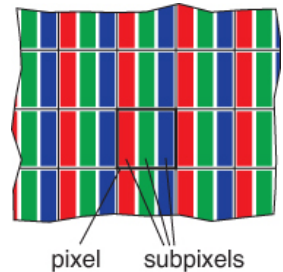
- ▶ Display
- ▶ Projector

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Common Raster Devices:

Output:

- ▶ Display
- ▶ Projector

Hardcopy:

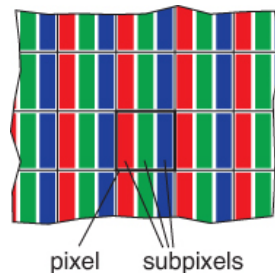
- ▶ ink-jet printer

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Most images are presented on a kind of *raster display*.

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- ▶ Red Subpixel
- ▶ Green Subpixel
- ▶ Blue Subpixel



Common Raster Devices:

Output:

- ▶ Display
- ▶ Projector

Hardcopy:

- ▶ ink-jet printer

Input:

- ▶ Digital camera
- ▶ Scanner

¹Pixel is short for "picture element"